INCH-POUND
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DETAIL SPECIFICATION

COVERALLS, RADIOACTIVE CONTAMINANTS PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of radioactive contaminants protective coveralls.

1.2 Classification. The coveralls are available in the following sizes:

Small/Medium
Large/X-Large

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be sent to trpsptctstandardiz@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

FEDERAL SPECIFICATIONS

JJ-W-155 - Webbing, Textile, (Cotton, Elastic)

COMMERCIAL ITEM DESCRIPTIONS

A-A-52094 - Thread, Cotton
A-A-55126 - Fastener Tapes, Hook and Loop, Synthetic

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1491 - Glossary of Knitting Imperfections

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

PURCHASE DESCRIPTIONS

NCTRF PD 01-11 - Cloth, Twill, Polyester and Cotton

DRAWINGS

15-NCTRF-22 - Symbol, Radiation

(Copies of these documents are available from the procuring activity. See 6.3.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC TM 20A - Test Method for Fiber Analysis: Quantitative

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(Copies of this document are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabric
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3786/D3786M - Standard Test Method for Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6193 - Standard Practice for Stitches and Seams
ASTM D8007 - Standard Test Method for Wale and Course Count for Weft Knitted Fabrics

(Copies of these documents are available online at https://www.astm.org.)

AEROSPACE INDUSTRIES ASSOCIATION

NASM 20230 - Grommets, Metallic, Plain and Rolled Rim, With Washer, Type I and Type III

(Copies of this document are available online at https://www.aia-aerospace.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection, in accordance with 4.2.

3.2 Design. The coveralls shall be one-piece with full length sleeves and knitted wristlets. The front left side of the coveralls shall have a breast pocket with a flap and a waist pocket with a flap. A nylon drawcord shall be channeled at the neck for closure. Elastic webbing shall be

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incorporated at the back waist, at the leg hems, and for foot stirrups. A black radiation symbol shall be stamped on the upper left back of the coveralls. The upper back area of the coveralls shall have a fastener tape hood anchor point.

3.2.1 Figures. Figures 1, 2, and 5 are furnished for information purposes only. Figures 3, 4, and 6 shall apply. When inconsistencies exist between the written specification and figures, the written specification shall govern.

3.3 Materials. The materials shall conform to the applicable specifications and standards as listed or required herein.

3.3.1 Basic material. The basic material shall be a crease-resistant polyester and cotton blend twill cloth, Navy Shade Yellow 3511 conforming to NCTRF PD 01-11. The technical back side of the fabric shall be used as the face and shall finish as the exposed surface of the coveralls. See 6.3.

3.3.2 Wristlets. The cotton, 1 x 1 rib knit, wristlets shall be bleached or unbleached (natural) and conform to the requirements in Table I when tested as specified in 4.4. The cotton yarn shall be carded. The 1 x 1 rib knit shall be seamless and double thickness. The use of elastic yarns in the knit shall be prohibited.

TABLE I. Wristlet physical requirements.

Characteristic	Requirement
Weight, finished wristlet, oz per linear yard (minimum)	0.84
Wales per inch (minimum)	16
Courses per inch (minimum)	21
Bursting strength, lbs. (minimum)	115
Elongation – wale direction, percent	75 (± 15)
Elongation – course direction, percent	300 (± 15)

3.4 Components. The components shall conform to the applicable specifications and standards as listed or required herein.

3.4.1 Braid. The drawcord for the top of the coveralls shall be braided of multifilament nylon and shall conform to the requirements specified in Table II when tested as specified in 4.4. The finished braid shall be not less than 3/16-inch in width and the color shall be white. The two ends of the drawcord shall be knotted together. The finished length of the drawcord shall be the circumference of the drawcord channel plus 12-inches (minimum) of exposed drawcord with knotted end. See Figure 3.

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TABLE II. Braid physical requirements.

Characteristic	Requirement
Yarn denier	840
Number of carriers	16
Ends per carrier	3
Picks per inch (minimum)	11

3.4.2 Fastener tapes. The hook fastener tape shall conform to Type II, Class 1, Style A of A-A-55126 and the loop fastener tape shall conform to Class 1, Style A of A-A-55126. White fastener tape shall be used in Small/Medium size coveralls and red fastener tape shall be used in Large/X-Large size coveralls to distinguish size identification. Fastener tapes shall be as specified in Table III. See 4.4.

TABLE III. Fastener tapes, requirements, and use (inches).

Location	Tape	Width	Length
Pocket flap	Hook	1 (±1/16)	4-1/2 (±1/16)
Pocket	Loop	1 (±1/16)	4-1/4 (±1/16)
Center back (hood anchor point)	Loop	2 (±1/16)	4 (±1/16)

3.4.3 Grommets and washers. The brass with nickel plated plain grommets and plain washers shall conform to Size 1 of NASM 20230. See 4.4.

3.4.4 Thread. The thread for all seaming and stitching shall be cotton conforming to Type I of A-A-52094, Tex 60, 3-ply. The thread color shall match the basic material in 3.3.1. See 4.4.

3.4.5 Webbing, elastic. The webbing shall be cotton, elastic, unbleached or bleached white conforming to the Types and Classes of JJ-W-l55 as specified in Table IV. As an alternate, a woven or braided elastic webbing of equivalent grade as applicable may be used. See 4.4.

TABLE IV. Elastic webbing location and requirements (inches).

Location	Type	Class	Width	Length
Back waist take-up	I	10	1-1/2 (± 1/16)	15 (± 1/4)
Foot stirrups	I	7	1 (± 1/16)	10-1/2 (±1/4)
Leg hem	II	1	5/16 (± 1/32)	11-1/2 (±1/4)

3.4.6 Radiation symbol. The radiation symbol shall be 3-3/4 inches in diameter conforming to the design and dimensions in Drawing No. 15-NCTRF-22. The marking for the radiation symbol shall be in accordance with Type IV, Class 9 of MIL-DTL-32075. See 4.4.

3.5 Labels.

3.5.1 Label, barcode. Each pair of coveralls shall be individually barcoded in accordance with

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MIL-DTL-32075, Type VIII, Class 17. The tags shall have a hole to accommodate a 1/8 (± 1/16) inch diameter cord or string, 10 (± 1/4) inches in length. The tag shall be attached to the coveralls' hanger loop with the cord or string and secured with a knot. As an alternate a tagging gun or similar device may be used. The barcode element shall be a 13-digit national stock number (NSN). There shall be a 12-digit Universal Product Code (UPC) assigned for all NSNs by the government. The initials "UPC" shall appear beneath code. The barcodes for NSN and UPC shall be a medium to high density, be clearly legible and readable by scanner and shall be located so that they are completely visible on the coveralls when it is folded and or packaged as specified. The label's location shall cause no damage to the coveralls. See 4.4.

3.5.2 Label, combination. Each pair of coveralls shall have a combination identification and instruction label conforming to Type VI, Class 15 of MIL-DTL-32075. The printing shall show colorfastness to laundering. See 4.4. The inscription shall be as follows:

COVERALLS, RADIOACTIVE CONTAMINANTS, PROTECTIVE
FIBER CONTENT: 65% POLYESTER/35% COTTON
CONTRACT NO.: SPE1C1-00-0-0000 (Example)
NAME OF CONTRACTOR: ABC COMPANY (Example)
NAME OF MANUFACTURER: (If other than contractor)

LAUNDER IN ACCORDANCE WITH PRESCRIBED REQUIREMENTS

3.5.3 Label, size. The size labels shall conform to Type VI, Class 2 of MIL-DTL-32075. See 4.4. The size labels shall read as follows:

SIZE: SMALL/MEDIUM
CHEST/HIP: UP TO 43-INCHES
STOCK NO.: 8415-00-000-0000 (EXAMPLE)

OR

SIZE: LARGE/X-LARGE
CHEST/HIP: 44 TO 48-INCHES
STOCK NO.: 8415-00-000-0000 (EXAMPLE)

3.6 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Construction.

3.7.1 Patterns. The Government will furnish a complete set of patterns which show size, grain lines, cut quantity, placement markings and notches for assembly. The Government's patterns shall be used to create the contractor's working patterns (see 6.3). Minor modifications are permitted to the contractor's working pattern where necessary when using automatic equipment

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or production processes. These modifications shall not alter the appearance, serviceability, or dimensional requirements cited in this specification. All pattern parts shall be cut in accordance with directional lines marked on the pattern which indicate the warp direction, unless otherwise specified. The patterns provide a seam allowance of 1/2-inch for all regular double lapped, double stitched seams.

3.7.1.1 Pattern parts. The coveralls shall be manufactured from the pattern parts in Table V.

TABLE V. Pattern parts.

Material	Specification nomenclature	Pattern abbreviation	Cut No.
Basic material (3.3.1)	Front – Back	82243-FRONTBACK	2
	Front (alternate) 1/	82243-FRONTALT	2
	Back (alternate) 1/	82243-BACKALT	2
	Sleeve, front	82243-FRONTSLV	2
	Sleeve, back	82243-BACKSLV	2
	Pocket	82243-POCKET	2
	Pocket flap	82243-PKTFLAP	2

1/ If the width of the material does not allow for economical cutting, the patterns which provide for a four-piece body, may be used.

3.7.2 Cutting. The cloth shall be cut in strict accordance with the government furnished patterns. All component parts shall be cut from one piece of material. The filling flush (wrong) side of the fabric shall be the face and shall finish as the exposed surface of the coveralls.

3.7.3 Shade and size marking. The component parts of the coveralls shall be marked, ticketed, or bundled to insure a uniform shade and size throughout the coveralls. The use of ink pad numbering machine, rubber stamp or pencil is allowed, provided the numbering shall not show on the outside of the coveralls and wherever possible, shall be covered by the seam allowance. Any method may be used except:

1. Corrosive metal devices.
2. Sew-on shade tickets.
3. Adhesive type tickets which discolor or adhere upon removal of ticket.

3.7.4 Manufacturing operations requirements. The coveralls shall be manufactured in accordance with operation requirements specified in Table VI. The contractor is not required to follow the exact sequence of operations provided the finished coveralls is identical to that produced by following the sequence as listed in Table VI. Minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability or appearance requirements cited in the specification. Piece placement shall have a ± 1/16-inch tolerance from the pattern placement marks unless otherwise specified. The abbreviations used in Table VI are as follows:

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BOB	-	Bobbin
Brtck	-	Bartack
LPR	-	Looper
NDL	-	Needle
SPI	-	Stitches per inch

3.7.5 Seams and stitching. The seam and stitch types specified in Table VI shall conform to ASTM D6193. When two or more methods for seam or stitch type are given for the same operation, any of them may be used. Where stitch type 401 is used, the looper (under thread) shall be on the inside of the coveralls. Ends of all seams produced by stitch type specified, all double needle seaming shall be 1/4-inch gauge with the rows of stitching sewn so that no raw edges, runoffs, twists, pleats, or open seams will result. All seams shall start and finish evenly. Thread tension shall be maintained so that there is no tight or loose stitching. Hook and loop fastener tape stitching shall be 1/16 to 1/8-inch from the edge of the tape. The bight of the overedge stitching shall be 3/16-inch. The guide and knife shall be set to trim only the raveled ends of the fabric.

3.7.5.1 Stitches per inch. The stitches per inch shall be as specified in Table VI.

3.7.5.2 Thread breaks and ends of seams. Ends of stitching not caught in another line of stitching shall be backstitched not less than 3/8-inch. The ends of the label stitching shall be overlapped not less than 3 stitches. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than 1/2-inch. Ends of all seams produced by stitch type 401 shall be caught in other seams or stitching. Unless otherwise specified, all double needle seaming shall be 1/4-inch gauge with the row of stitching next to the folded edge not more than 1/16-inch from edge. The seam allowance shall be maintained with seams sewn so that no raw edges, runoffs, twist, pleat, or open seam will result. When stitch type 401 is permitted as an option, it shall not be used unless the ends are caught in other seams or stitching. Edge stitching and top stitching shall be 1/16-inch from folded edge.

TABLE VI. Table of operations.

No.	Operation	Stitch Type	Seam Type	SPI	Thread	
					NDL	BOB/ LPR
1.	Pockets, breast and waist. (See Figure 6)					
	a. Overedge pocket piece on all four sides.	503 or 504 or 602	EFd-1	8-10	50	50
	b. Position 1-inch-wide loop fastener tape 7/8-inch from top raw edge of pocket according to pattern marks. This forms the top pocket hem. Turn hem under. The top edge of the tape shall be flush against the finished top pocket edge and its length shall extend the full width of the finished pocket.	301	LSbj-1	10-12	30	30

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TABLE VI. Table of operations - Continued.

No.	Operation	Stitch Type	Seam Type	SPI	Thread	
					NDL	BOB/ LPR
1. cont.	The tape shall be stitched 1/16 to 1/8-inch from edge on all four sides.					
	c. Position pockets on front according to marks on patterns. The pockets shall be placed within 1/4-inch or less from the placement markings on patterns. Fold bottom and sides of pocket under 3/8-inch. Stitch down sides and bottom 1/16-inch from folded edges.	301	LSd-1	10-12	30	30
2.	Pocket flaps, breast and waist. (See Figure 6)					
	a. Fold flap, face-to-face, and stitch sides 3/8-inch from edge. Turn to finished position and work out points.	301	SSa-1	10-12	30	30
	b. The 1-inch-wide hook fastener tape shall be positioned on underside of finished flaps according to pattern marks. The tape shall be placed against the finished flap edge and its length shall extend the full width of the finished flap. The tape shall be stitched 1/16 to 1/8-inch from edge on all four sides.	301	LSbj-1	10-12	30	30
	c. Stitch the underside of one pocket flap to left front chest and a second pocket flap to the left front waist, 1-inch above the finished pockets according to pattern marks. The pocket flap shall have a 3/8-inch seam allowance/fold line that aligns with the pocket flap line marked on front panel pattern. The pocket flap shall be placed within 1/4-inch or less from the placement markings on patterns. Turn down to finished position and stitch 1/4-inch from folded edge.	301	LSs-1	10-12	30	30
3.	Front-Back.					
	a. Double stitch center seams of fronts and backs with lefts lapped over rights (as worn).	301 or 401	LSc-2	10-12	30 50	30 50
	b. For 4-piece construction: Double stitch right and left fronts to right and left backs respectively with fronts lapped over backs (as worn).	301 or 401	LSc-2	10-12	30 50	30 50

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TABLE VI. Table of operations - Continued.

No.	Operation	Stitch Type	Seam Type	SPI	Thread	
					NDL	BOB/ LPR
3. cont.	c. Position 15-inch elastic webbing on backs to correspond to marks on pattern and stitch with 4 equidistant rows of stitching with the outer rows stitched 1/16 to 1/8-inch from outer edge along elastic webbing. When under tension the elastic webbing shall stretch to 24-inches long (minimum). Stitch ends of elastic webbing 1/16 to 1/8-inch from each edge.	301	SSau-4	10-12	30	30
4.	Sleeves.					
	a. Double stitch the top seam and bottom seam of sleeve with the fronts lapped over the backs.	301 or 401	LSc-2	10-12	30 50	30 50
	b. Cut wristlet 8-1/2 inches long and fold in half lengthwise. Overedge stitch the wristlet to the end of the sleeve.	503 or 504 or 602	SSa-1	8-10	50	50
5.	Armholes.					
	Double stitch the sleeves to armholes with body lapped over sleeves.	301 or 401	LSc-2	10-12	30 50	30 50
6.	Inseams.					
	Double stitch the leg inseams in one continuous operation with the fronts lapped over the backs.	301 or 401	LSc-2	10-12	30 50	30 50
7.	Drawcord channel – top hem.					
	a. Secure grommet and washer 1-1/2 (± 1/16) inches from top raw edge, centered on the center front seam.					
	b. Turn under 1/4-inch top edge, fold over to make 3/4 (± 1/8) inch wide hem and stitch 1/16-inch from bottom folded edge, inserting size label approximately centered over the back joining seam.	301	EFb-1	10-12	30	30
	c. Insert drawcord into hem (drawcord channel). Sear and knot the ends to prevent raveling and loss of free end of cord in the channel. See Figure 3.					

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TABLE VI. Table of operations - Continued.

No.	Operation	Stitch Type	Seam Type	SPI	Thread	
					NDL	BOB/ LPR
8.	Hanger loop. (See Figure 5)					
	a. The loop shall be made by folding a piece of the basic material lengthwise, so the edges are abutting in the center. Stitch with a cover stitch on the underside. OR b. Fold hanger stripping in half with the raw edges turned in and stitch 1/16 to 1/8-inch from each edge.	406 301	EFh-1 EFp-2	10-12 10-12	50 30	50 30
	c. Position the hanger loop lying flat on the inside back of the coveralls, centered over the center back seam. Fold under each raw edge of the hanger loop 1/4-inch and stitch each side with a 3/8-inch vertical bartack simultaneously catching the channeled nylon drawcord.	Brtck		28 per brtck	50	50
9.	Elastic loop - leg hem.					
	a. Cut elastic webbing 11-1/2 inches long. Place one end of the elastic webbing overlapping the other end 1/2-inch and backstitch or bartack across the width to form a continuous loop 11-inches in circumference.	301 or Brtck	LSa-1	10-12 28 per brtck	30 50	30 50
	b. Mark a 3/4 (± 1/8) inch line and turn up on each leg bottom with the raw edge turned under 1/4 to 3/8-inch. Stitch 1/16-inch from inside folded edge simultaneously inserting the formed elastic loop without catching it in the stitching.	301	EFb-1	10-12	30	30
10.	Foot stirrup straps.					
	Position one end of the 10-1/2 inch elastic webbing adjacent to inside leg hem inseam and the other end on the opposite side of leg. Stitch two 3/8-inch long bartacks at each end of strap, along hem seam line, 1/8 to 1/4-inch from raw edge.	Brtck		28 per brtck	30	30

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TABLE VI. Table of operations - Continued.

No.	Operation	Stitch Type	Seam Type	SPI	Thread	
					NDL	BOB/ LPR
11.	Hood anchor point.					
	Position a loop fastener tape to center back of coveralls as indicated by marks on patterns. Stitch 1/16 to 1/8-inch from edge on all four sides of fastener tape.	301	LSbj-1	10-12	30	30
12.	Labels. (See Figure 5)					
	a. Layer the combination label and size label top edges and position on the inside of the coveralls adjacent to the folded edge of drawcord channel hem and centered over the back joining seam.					
	b. Stitch the combination label on all four sides 1/16-inch from edge. Stitching shall not go through printing. Top edge of the combination label and size label may be caught when performing operation 7b.	301	LSbj-1	10-12	30	30
13.	Radiation symbol. (See Figure 4)					
	Stamp radiation symbol, as indicated in Drawing No. 15-NCTRF-22 on the upper left back section of the coveralls midway between the center back and sleeve-armhole seams with the bottom of the symbol aligned with the base of the armhole.					
14.	Cleaning.					
	a. Trim all thread ends. Care shall be taken when trimming thread ends that damage to the fabric does not result.					
	b. Remove all loose threads.					
	c. Remove all spots and stains.					

3.7.6 Replacement of defective components. During the spreading, cutting, and manufacturing process, component parts having material defects or damages that are classified as defects in section 4 shall be removed from production and replaced with non-defective and properly matched components.

3.8 Finished dimensions. The finished dimensions of the coveralls shall be in accordance with Table VII. See 4.7.1.

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TABLE VII. Finished dimensions, (inches).

	Small/Medium	Large/X-Large	Tolerance
Chest (A)	53	58	± 1
Seat (B)	54	58	± 1
Leg inseam (C)	31	33	± 1/2
Sleeve inseam (D)	25	25	± 1/2
Back length (E)	64-1/2	70	± 1/2
Knitted cuff length (F)	4		± 1/4
Knitted cuff width (G)	3		± 1/4
Hanger loop width (H)	1/4		+ 1/8, – 0
Hanger loop free opening (J)	2-1/2		+ 1/4, – 0
Pocket width (K)	4-1/4		± 1/8
Pocket height (L)	3-3/4		± 1/8
Pocket flap width (M)	4-1/2		± 1/8
Pocket flap height (N)	2-1/8		± 1/8

3.9 Workmanship. The finished coveralls shall conform to the quality established in this specification. The coveralls shall be uniform in quality and shall be free from irregularities or defects affecting appearance, performance, reliability, or durability. The occurrence of defects cited in Table VIII shall not exceed the acceptance criteria established. On the same coveralls the following shall apply:

TABLE VIII. Variation allowance, (inches).

Coveralls location	Maximum variation allowance
Left and right sleeve, even in length	1/2
Left and right leg, even in length	1/2
Left and right wristlet, even in length and width dimensions	1/4
Pockets and flaps, horizontal alignment	1/4
Two pocket/flap assemblies, vertical alignment	1/4
Crotch seam staggering	1/2
Hanger loop centering from center back	3/4

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article inspection shall consist of the examinations and tests specified in 4.4 through 4.7.1. The coveralls shall be inspected for compliance with design,

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construction, workmanship, appearance, and dimensional requirements specified in this document. If the requirements specified in this document are not met, the first article inspection shall be unacceptable.

TABLE IX. First article inspection verification.

Inspection	Paragraphs	
	Requirement	Verification
Materials and components examinations	3.3 through 3.5.3	4.4
Visual examination	3.7.4 through 3.7.6	4.7
Dimensional examination	3.8	4.7.1
End item testing	3.9	4.7 through 4.7.1

4.2.1 First article samples. Unless other specified, first article samples shall be provided. The sample size will be specified in the procurement document (see 6.2). The sample unit shall be one finished pair of coveralls. The lot size shall be expressed in units of coveralls.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in Section 4.

4.3.1 Conformance inspection sample. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4. The sample unit shall be one finished pair of coveralls. The lot size shall be expressed in units of coveralls.

4.3.2 Component and material inspection. In accordance with 4.3, components and materials shall be examined and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4 Component testing. The components specified in 3.3 through 3.5.3 shall be tested for the characteristics and test methods listed in Table X. All test reports shall contain the individual values utilized in expressing the final results.

TABLE X. Component and material verification.

Material	Characteristics	Requirement	Verification 1/
Basic material	All	3.3.1	NCTRF PD 01-11
Wristlet	Fiber identification	3.3.2	AATCC TM 20A
	Material structure		Visual 2/
	Fabric weight		ASTM D3776/D3776M, Opt. C
	Wales and courses per inch		ASTM D8007
	Bursting strength		ASTM D3786/D3786M
	Elongation, wale and course direction		ASTM D5034

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TABLE X. Component and material verification - Continued.

Material	Characteristics	Requirement	Verification 1/
Wristlet, cont.	Width	3.3.2	Visual 2/
	Color		Visual 2/
Braid	Fiber identification	3.4.1	AATCC TM 20A
	Denier		ASTM D1907/D1907M
	Material structure		Visual 2/
	Number of carrier yarns		ASTM D3775
	Number of ends per carrier		ASTM D3775
	Picks per inch		ASTM D3775
	Width		Visual 2/
	Color		Visual 2/
Fastener tapes	All	3.4.2	A-A-55126
Grommets & washers	All	3.4.3	NASM 20230
Thread	All	3.4.4	A-A-52094
Webbing, elastic	All	3.4.5	JJ-W-155
Radiation symbol	All	3.4.6	MIL-DTL-32075
	All		15-NCTRF-22
Labels	All	3.5	MIL-DTL-32075

1/ When end item document is cited, all requirements and test methods listed in applicable end item document shall be performed.
2/ One wristlet or one 6-inch-long braid shall be examined for the characteristics cited in Table X. The results shall be recorded as pass or fail.

4.5 In-process inspection. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details which cannot be examined in the finished product are in accordance with requirements specified in Section 3. Materials and components, which can be classified as a defect in accordance with Table XI, shall be removed from production.

4.6 Inspection methods. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in Section 4 of this specification.

4.7 End item visual examination. The coveralls shall be examined for defects listed in Table XI.

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TABLE XI. Classification of defects.

Examination	Defect	Classification	
		Major	Minor
Material and workmanship	Any weakening material defect, such as a smash, hole, cut, tear, burn, loose slub, mend, thin place, needle chew or drill hole	101	
	One or more broken or missing yarns	102	
	Any non-weakening material defect such as knot, mis-weave, unsightly slub, loose yarn, coarse yarn, area of no dye penetration, dye streak or woven in waste		201
Cleanness	Any spot or stain on outside		202
	Thread ends not trimmed to 1/4-inch or loose threads, lint, and foreign matter not removed (to be scored only when condition exists on major portion of coveralls)		203
	Two or more shade or size tickets not removed		204
	Shade marking exposed and visible on outside of coveralls		205
Shade	Shaded parts	103	
	Color of component not as specified	104	
	Thread color not matching basic material	105	
Cutting	Any part not cut in accordance with directional lines indicated on patterns or in accordance with the specification requirements	106	
Component and assembly	Any component part omitted or not as specified (type, size, etc.); any operation omitted or not as specified (unless otherwise classified herein)	107	
	Any component part twisted, distorted, pleated, tight, or full		206
Seams and stitching	No open seams, raw edges, runoff, skipped or broken stitches	108	
	Seam puckered, irregular, wavy, or pleated (unless otherwise classified herein)		207
	Part of coveralls caught in any unrelated stitching operation	109	

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TABLE XI. Classification of defects - Continued.

Examination	Defect	Classification	
		Major	Minor
Seams and stitching, cont'd	Permanent fold or pleat in cloth (caught in seaming or stitching)		208
	End of stitching when not caught in another seam or stitching backstitched less than 3/8 inch		209
	Ends of a continuous line of stitching overlapped less than 1/2-inch		210
	Needle perforations visible on outside		211
	Thread breaks not secured or stitched back of the break less than 1/2-inch	110	
	Gage of stitching irregular, not uniform or not within range of specified by 1/16-inch or more (score when condition occurs on more than half the length of seam)		212
	Edge of top stitching sewn too close to edge, resulting in damaged cloth		213
	Width of seam allowance not as specified by 1/16-inch or more than half the length of the seam		214
	Seam type or stitch type not as specified	111	
	Looper thread of 401 stitch type finishing on the outside of the coveralls	112	
	Loose tension resulting in loose seam		215
	Loose tension (exposed loops of top or bottom threads)		216
	Tight tension (stitches break when normal strain is applied in direction of the seam or stitching)		217
	Stitches per inch not as specified (score only when condition exists on more than half the length of the seam)		218
	Any seam not lapped in the direction specified		219
	Crotch seam and seat seam staggered more than 1/2-inch (measured from center to center of seams)		220

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TABLE XI. Classification of defects - Continued.

Examination	Defect	Classification	
		Major	Minor
Cloth	Face side of cloth not as specified	113	
Drawcord	Omitted; drawcord in drawcord channel not caught under hanger loop bartacks	114	
	Free end of drawcord less than 10-inches in length when neck opening fully (drawcord channel) is extended		121
Drawcord channel	Channel hem width not as specified	115	
	Cord ends not knotted to form a continuous loop	116	
	Drawcord caught in hem stitching	117	
Elastic, back waist	Other than four rows of stitching		222
	Not stitched under tension to 24-inch length		223
	Width or length not as specified	118	
Elastic, foot stirrup	Bartack missing, insecure, misplaced, width or length not as specified	119	
Elastic, hem	Loop not continuous	120	
	Loop caught in hem stitching		224
	Half measurement less than 5-1/2 inches when hem laid flat and relaxed	121	
	Hem not width as specified.	122	
Fastener tape	Misplaced; type, length, width, or color not as specified	123	
	Foreign matter (thread, etc.) trapped in hook tape		225
	Less than 25 percent of the hooks or loops (pile) flattened or missing from any fastener tape closure area.		226
	25 percent or more of the hooks or loops (pile) flattened or missing from any fastener tape closure area.	124	
	Hook and loop fastener tape incapable of engaging properly when pocket is in use.	125	
Grommet	Broken, not securely clinched to fabric, too tightly clinched cutting fabric	126	
	Any sharp or rough edge		227
	Not on center front seam		228
	Does not provide clear access into hem	127	

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TABLE XI. Classification of defects - Continued.

Examination	Defect	Classification	
		Major	Minor
Hanger loop	Off-center from center back seam by 3/4-inch		229
	Opening not as specified; width not as specified		230
Knitted wristlet	Type or dimensions not as specified	128	
	Not double thickness	129	
	Left and right length and width dimensions vary from each other more than 1/4-inch		231
	Cloth not 1 x 1 rib as specified	130	
	Elastic yarn present in knit wristlet	131	
	Any knit defect cited under MIL-STD-1491.	132	
Labels	Omitted, incorrect or illegible	133	
	Misplaced		232
	Stitching through printing		233
Label, barcode	Barcode omitted or not readable by scanner; human readable interpretation (HRI) omitted or illegible; not visible on folded, packaged coveralls		234
	Barcode cord or string omitted, misplaced, dimensions not as specified; Barcode label not visible when folded and/or packaged; causing damage to coveralls	134	
Pockets	Not positioned properly by 1/4-inch or more	135	
	Less than 4-1/8 inches wide or more than 4-3/8 inches wide	136	
	Less than 3-5/8 inches long or more than 3-7/8 inches long	137	
Pocket flaps	Not positioned properly by 1/4-inch or more	138	
	Less than 2-1/8 inches long	139	
	More than 4-5/8 inches wide	140	
Radiation symbol	Omitted, illegible, misplaced, or not as specified	141	
Finished appearance	Sleeve or leg inseam length uneven by more than 1/2-inch.	142	
	Sleeve pleated or puckered at armhole		235

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4.7.1 Dimensional examination. The coveralls shall be examined for conformance to the dimensions specified in Table VII. The measurements shall be taken with the coveralls laid flat on a smooth, flat surface and the neck opening fully extended. See Figures 3 through 6. When finished dimensions are out of tolerance, they shall be scored as Major Defect 143. The finished coveralls shall be measured as follows:

A. Chest. Twice the measurement taken across the front, at the base of the armholes, from folded edge to folded edge.

B. Seat. Twice the measurement taken across the front, 6-inches above the crotch seam, from folded edge to folded edge.

C. Leg inseam. Measurement shall be taken from the crotch point, along inseam, to the bottom edge of hem.

D. Sleeve inseam. Measurement shall be taken from the sleeve joining seam, along inseam, to the bottom edge of the knitted cuff.

E. Back length. Measurement shall be taken 4-inches from the center seam, from top edge of neck opening to bottom edge of leg hem.

F. Knitted cuff length. Measurement shall be taken from the sleeve-cuff joining seam to the bottom edge of the cuff, on the outside of coveralls.

G. Knitted cuff width. Measurement shall be taken from cuff folded edge to folded edge.

H. Hanger loop width. Measurement shall be taken from top finished edge to bottom finished edge through the center of the hanger loop.

J. Hanger loop free opening. Measurement shall be taken between bartacks through the center of the hanger loop.

K. Pocket width. Measurement shall be taken from one side finished edge to the other side finished edge through the center of the pocket.

L. Pocket height. Measurement shall be taken from top finished edge to bottom finished edge through the center of the pocket.

M. Pocket flap width. Measurement shall be taken from one side finished edge to the other side finished edge through the center of the flap.

N. Pocket flap height. Measurement shall be taken from top finished edge to bottom finished edge through the center of the flap.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The radioactive contaminants protective coveralls are worn by the Naval Nuclear Propulsion community.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including modifications.
b. Size and quantity (see 1.2).
c. National Stock Numbers.
d. Applicable Government patterns, drawings, including revisions.
e. First article and conformance inspection requirements (3.1).
f. Number of first article inspection samples (4.2.1).
g. Name and address of first article inspection facility; and the name and address of the Government activity responsible for conducting the first article inspection program.
h. Inspection conditions
i. Conformance inspection and acceptance quality limits
j. Packaging required (see 5.1).

6.3 Information. For access to information such as specification, patterns, drawings, shade standard samples of cloth, go to DLA Troop Support's Clothing and Textile website for their Specification/Drawing/Pattern Request Form: https://www.dla.mil/TroopSupport/ClothingandTextiles/SpecRequest.aspx. Complete the request form and then submit. Requests to use equivalent materials and/or components or to make changes to the pattern should be sent to the contracting officer for approval by the military services.

6.4 Subject term (key word) listing.

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – NU
Air Force – 11
DLA – CT

Preparing activity:
DLA – CT
Project (8415-2025-014)

Review Activities:
Navy – AS, OS, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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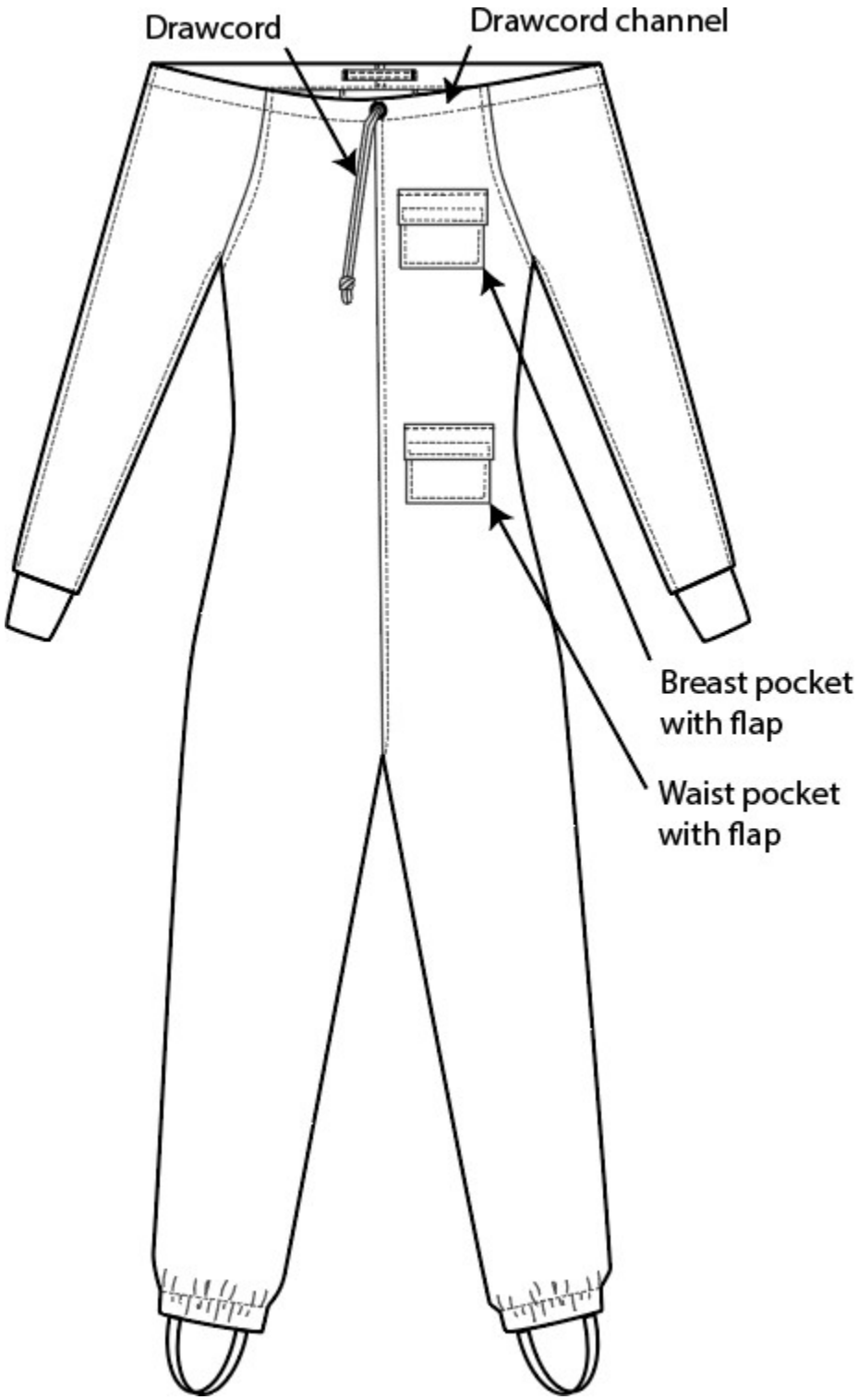


FIGURE 1. Coveralls details, front view.

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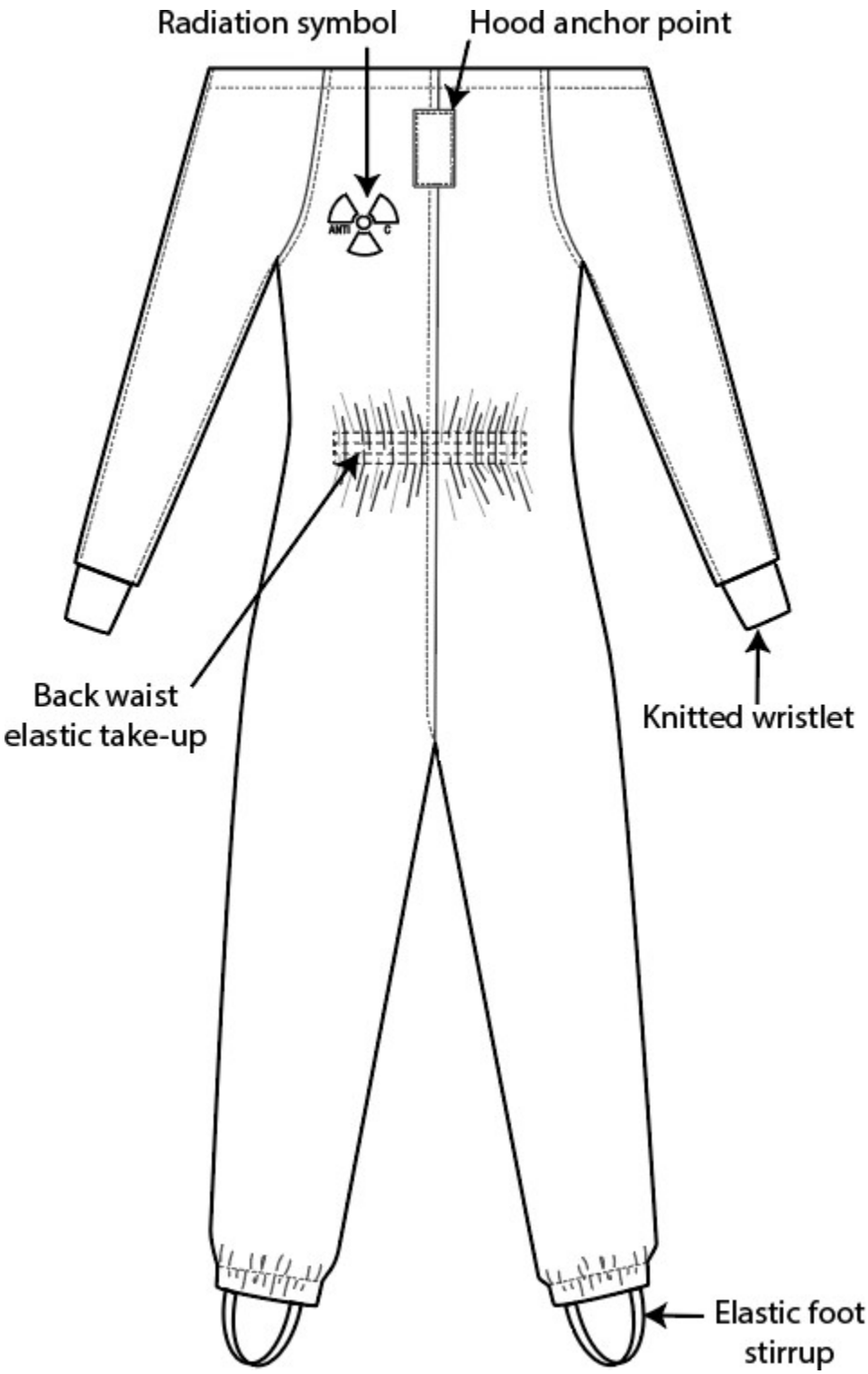
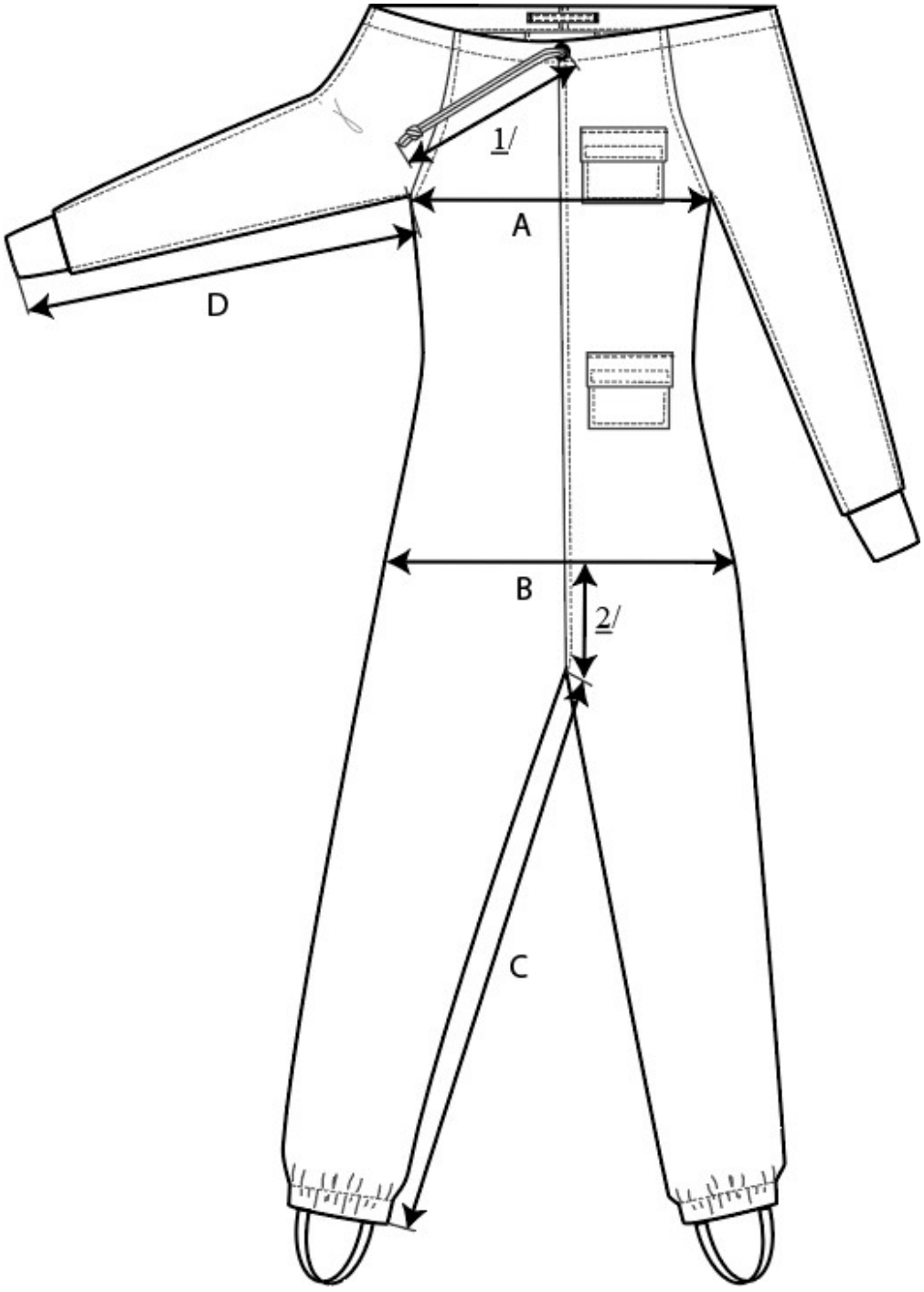


FIGURE 2. Coveralls details, back view.

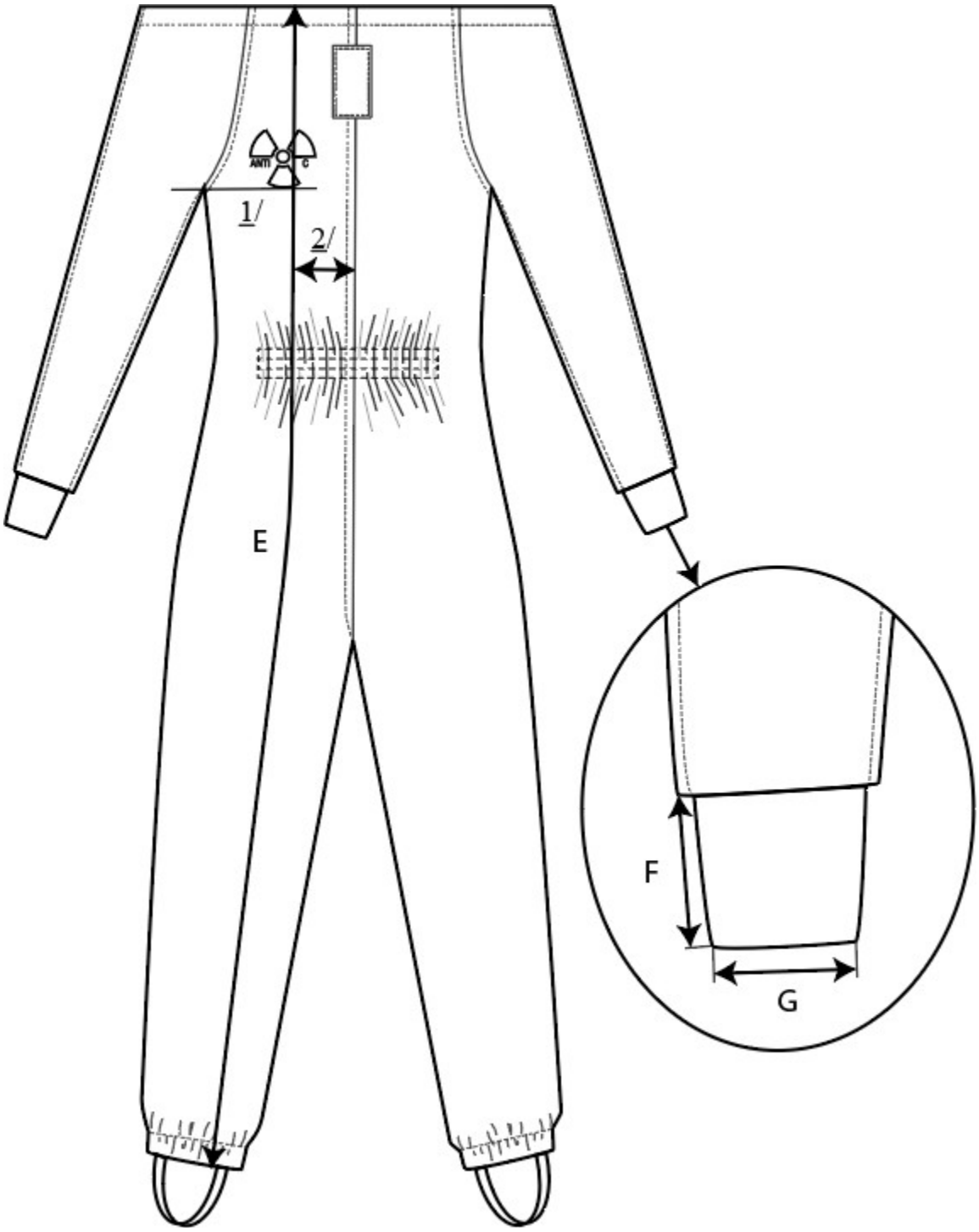
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NOTES:
1. Drawcord free end length, see 3.4.1
2. Seat measurement, see 4.7.1

FIGURE 3. Coveralls measurement locations, view.

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NOTES:
1. Radiation symbol placement, see Table VI, Operation 13.
2. Back length measurement, see 4.7.1

FIGURE 4. Coveralls measurement locations, back view.

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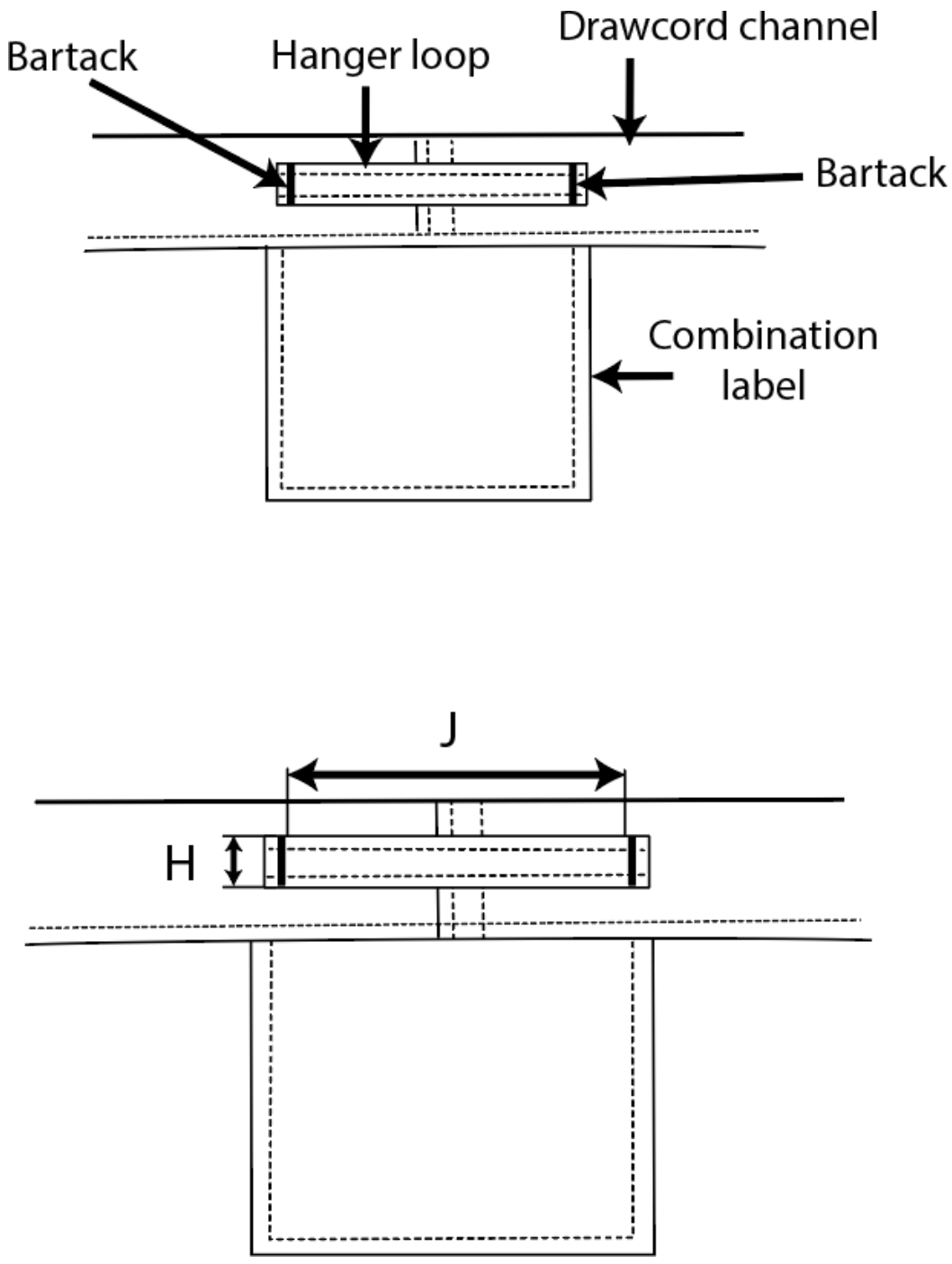
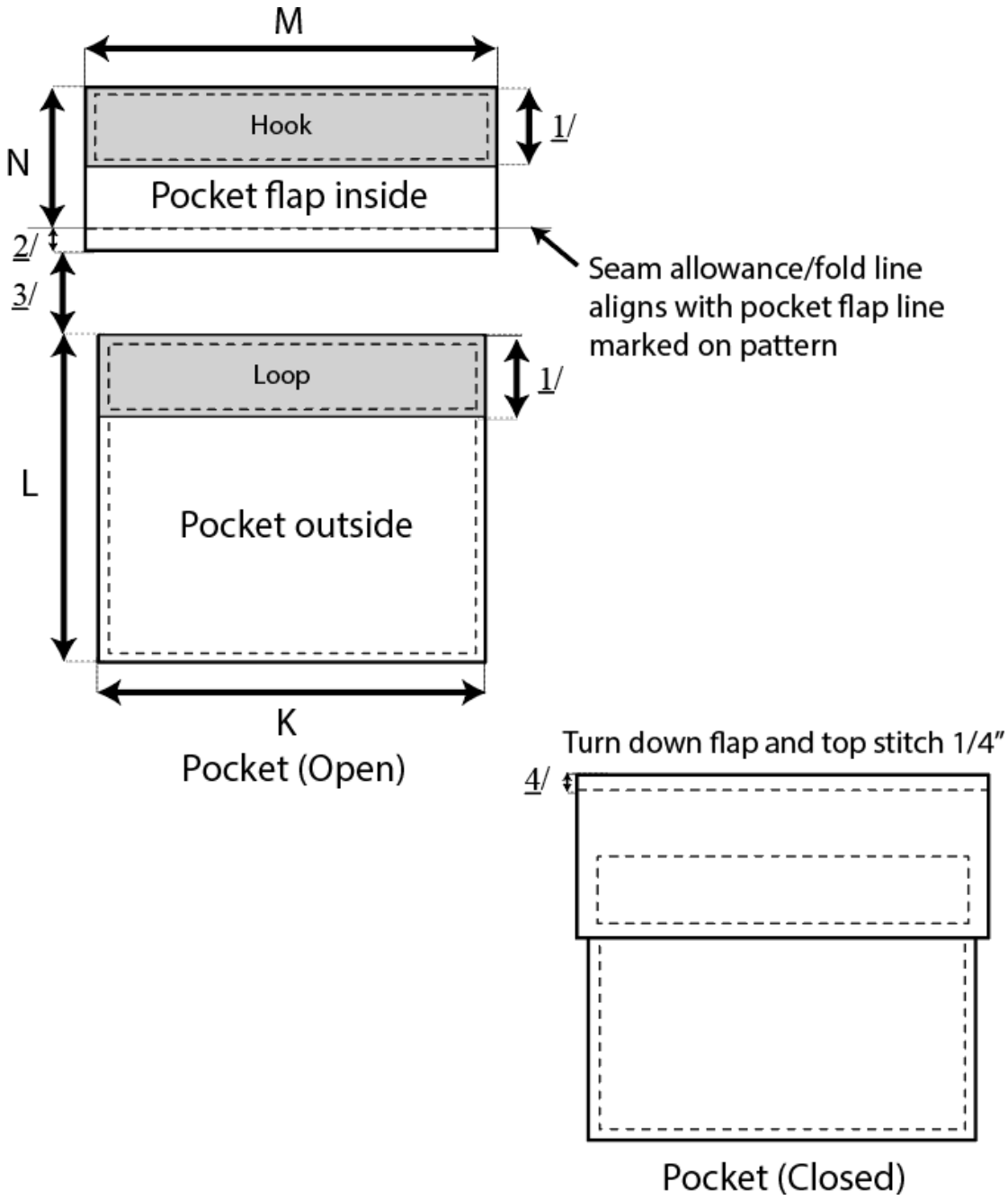


FIGURE 5. Hanger loop, detail views.

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NOTES:
1. Hook and loop fastener tape width, see 3.4.2, Table III
2. Distance from seam allowance/fold line to finished edge of pocket flap, see Table VI, Operation 2c.
3. Distance from finished edge of pocket flap to finished edge of pocket, see Table VI, Operation 2c.
4. Topstitching, see Table VI, Operation 2c.

FIGURE 6. Pocket and pocket flap, detail views.

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